COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE007	**SECTION:**	CPE11S4
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ACTIVITY NO. 1.2 Writing Your First Program using C+

1. PROCEDURE OUTPUT

1. Try to create a simple program using C++ language that outputs your whole name. Using the new line syntax, output your program and the course and section



EXPLANATION: This is a C++ program that outputs my name, program, course, and section under each other. To accomplish this task, I used the endl syntax wherein it puts the next out put in a new line since I am ending the previous line.

2. Find all possible compilation errors and logic errors. Fix them. Your version of the program must print the same result as the expected output. Before you use your compiler, try to find the errors only by manual code analysis.

```
#include <iostream>
        int main()
        {
        cout("The value of five is:"<< 5int);
        return 0;
        }
```

1. There was integration of the using namespace std which is important for us to use the C++ language.
2. There was a parenthesis between the cout, we primarily use parenthesis in operations, not in outputs.
3. The << symbol is missing after the cout.
4. There should be no int beside the 5.

main.cpp

Share Run Output

```cpp
#include <iostream>
using namespace std;

int main()
{
    cout << "The value of five is:" << 5;
    return 0;
}
```

```
The value of five is:5

=== Code Execution Successful ===
```

EXPLANATION: This is the fixed version of the given code whereas this code functions properly and outputs the intended number, after identifying the errors manually, I was able to narrow down and repurpose the given code into something that works properly.

3. Design an algorithm that reads the base and height of a triangle and displays its area

main.cpp

Share Run Output

```cpp
//Design an algorithm that reads the base and height of a triangle and displays its area
#include <iostream>
using namespace std;

int main () {
    double base, height, area;
    cout << "INPUT A BASE: ";
    cin >> base;
    cout <<"INPUT A HEIGHT: ";
    cin>> height;

    area = 0.5*base*height;
    cout<< "AREA IS: " << area;

    return 0;
}
```

```
INPUT A BASE: 5
INPUT A HEIGHT: 7
AREA IS: 17.5

=== Code Execution Successful ===
```

EXPLANATION: This is a simple program that follows the given situation wherein the user inputs a base and a height which will then be calculated for its area using the formula ½ base x height. First is to declare the variables, and in order to get the input from the user, I used the cin syntax, and for the program to calculate its area, I used a series of operations that corresponds to the formula for the area of a triangle. Lastly is to output the calculated value from the variable area by using cout.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. Design an algorithm that reads the midterm grade, final grade, and laboratory grade of a student and displays the average grade

main.cpp

Share Run

```cpp
//Design an algorithm that reads the midterm grade, final grade, and laboratory grade of a student and
//displays the average grade
#include <iostream>
using namespace std;

int main () {
    double MG, FG, LG, sum, ave;
    cout << "Input your Midterm Grade: ";
    cin >> MG;
    cout << "Input your Final Grade: ";
    cin >> FG;
    cout << "Input your Laboratory Grade: ";
    cin >> LG;

    sum = MG+FG+LG;
    ave = sum/3;

    cout <<endl;
    cout << "Your Average Grade is: " << ave;


    return 0;
}
```

Output

```
Input your Midterm Grade: 95
Input your Final Grade: 96
Input your Laboratory Grade: 99

Your Average Grade is: 96.6667

=== Code Execution Successful ===
```

EXPLANATION: The first supplementary activity made us do a program that reads the midterm, final, and laboratory grade of the user. To start the program I first declared variables that would be needed to collect the inputs of the user. Next is the process, to get the average of the grades I initially declared a variable for sum which in return calculates the sum of the grades first before performing the division in the average variable to make the code simpler and easier to follow. Since the scenario is only looking for the average grade, the only output I used cout with is the average variable.

2. Design an algorithm that reads the name of a product, unit price, and quantity purchased, then displays the product name, quantity, and total amount to pay

main.cpp

```cpp
//Design an algorithm that reads the name of a product, unit price, and quantity purchased, then displays the product name, quantity, and total amount to pay.
#include <iostream>
using namespace std;
#include <string>

int main () {
    string prodname;
    double up, quant, amtp, sisig;

    cout << "Welcome to Cusineros!"<<endl;
    cout << "Your order is: ";
    cin >> prodname;
    cout << "Unit Price: PHP ";
    cin >> up;
    cout << "Quantity Purchased: ";
    cin >> quant;

    amtp = up*quant;

    cout<<endl;
    cout<< "THANK YOU!" << endl;
    cout <<" You ordered: "<< prodname<< endl;
    cout <<" Quantity: "<< quant << endl;
    cout <<" Amount to pay: PHP "<< amtp << endl;

    return 0;
}
```

Output

```
Welcome to Cusineros!
Your order is: Sisig
Unit Price: PHP 79
Quantity Purchased: 5

THANK YOU!
 You ordered: Sisig
 Quantity: 5
 Amount to pay: PHP 395


=== Code Execution Successful ===
```

EXPLANATION: This program asks the user to enter the product name, unit price, and quantity purchased. It stores the information entered by the user, then calculates the total amount by multiplying the unit price by the quantity. After the calculation, the program displays the product name, quantity purchased, and the total amount to pay. This helps the user easily see the complete details of the purchase.

3. Design an algorithm that reads a person's birth year and the current year and displays the person's age.

main.cpp

```cpp
//Design an algorithm that reads a person's birth year and the current year and displays the person's age.
#include <iostream>
using namespace std;

int main () {
    int byear, cyear, age;
    cout << "Hello! Kailan ka ipinanganak?" << endl;
    cout << "Birth Year: ";
    cin >> byear;
    cout << "Mind telling me anong year na ngayon?" <<endl;
    cout << "Current Year: ";
    cin >> cyear;

    age = cyear-byear;

    cout<<endl;
    cout << "Your age should be: " << age;
    cout <<" years old.";

    return 0;
}
```

Output

```
Hello! Kailan ka ipinanganak?
Birth Year: 2008
Mind telling me anong year na ngayon?
Current Year: 2026

Your age should be: 18 years old.

=== Code Execution Successful ===
```

EXPLANATION: This code, according to the given instruction, collects the user's birth year and current year which then calculates the age of the user through a series of operations. To start the code, I first declared the variables and used cin syntax to gather data from the user which then processes it and transforms in the output itself.

4. Design an algorithm that reads the hourly wage and the number of hours worked and displays the gross salary.

main.cpp

Share Run

```cpp
//Design an algorithm that reads the hourly wage and the number of hours worked and displays the gross
//salary.
#include <iostream>
using namespace std;

int main () {
    double hw, wh, gslry, wdpm;
    wdpm = 20;


    cout <<"Hello Worker!"<<endl;
    cout << "Your hourly wage: PHP ";
    cin >> hw;
    cout << "Your work hours: ";
    cin >> wh;

    gslry = hw*wh*wdpm;

    cout<<endl;
    cout << "Your gross salary for a month is: PHP "<< gslry;


    return 0;
}
```

Output

```
Hello Worker!
Your hourly wage: PHP 500
Your work hours: 8

Your gross salary for a month is: PHP 80000

=== Code Execution Successful ===
```

EXPLANATION: In this program, I ask the user to enter the hourly wage and the number of hours worked. Then, I calculate the gross monthly salary by multiplying the hourly wage, work hours, and the fixed number of working days in a month. Finally, I display the total gross salary for the month.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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